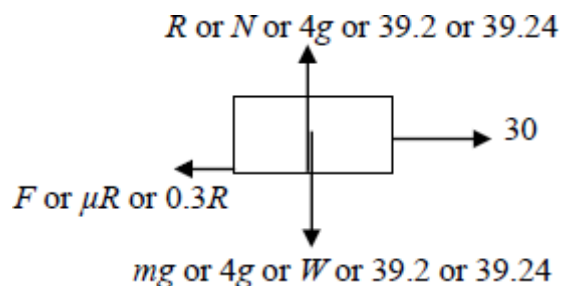


Mark schemes

Q1.

(a)



B1: Diagram with four forces showing arrow heads and labelled.

Ignore negative signs in labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 4 kg for the weight.

Note Accept μR for F .

B1

1

(b) $(R = 4 \times 9.8 =) 39.2 \text{ N}$

B1: Correct normal reaction. Accept 4g

B1

1

(c) $(F =) 0.3 \times 39.2 = 11.76 = 11.8 \text{ N (to 3sf)}$

M1: Use of $(F =) \mu R$

M1

A1: Correct friction.

Accept 1.2g or 11.7 or 11.76 N.

Do not condone further work after the value for friction has been obtained.

A1

2

(d) $4a = 30 - 11.76$

M1: Three term equation of motion.

A1F: Correct equation.

M1A1F

$$a = \frac{30 - 11.76}{4} = 4.56 \text{ ms}^{-2}$$

A1F: Correct acceleration.
FT candidates F from part (c).
Accept 4.55 from 11.8.

A1F

3

[7]

Q2.

(a) $5g - T = 5a$

M1: Three term equation of motion with 5g or 49,
5a (not 5ga) and T.
A1: Correct equation.

M1A1

$T - 3g = 3a$

M1: Three term equation of motion with 3g or 29.4,
3a (not 3ga) and T.
A1: Correct equation.

M1A1

$2g = 8a$

$a \left(= \frac{2g}{8} \right) = 2.45 \text{ ms}^{-2}$

AG

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently
use signs in the opposite direction throughout, provided
they then give their final answer as 2.45. If the
final answer is -2.45 don't award the final A1 mark.

Special Case:

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Whole String Method $2g = 8a$ and $a = \frac{2g}{8} = 2.45$
OE M1A1A1.

5

(b) $T = 3 \times 9.8 + 3 \times 2.45$

M1: Substitution of $a = 2.45$ into a three term
equation of motion to find the tension. Contains T,
mg and ma where $m = 3$ or 5

M1

$= 36.75$

$= 36.8 \text{ N (to 3 sf)}$

A1: Correct tension.
Accept 36.75 or 36.7

A1

2

(c) Light and Inextensible

B1: Light

B1: Inextensible (Allow inelastic or not stretchy)

Ignore irrelevant non-contradictory assumptions.

B1B1

2

(d) (i) $0.196 = \frac{1}{2} \times 2.45 \times t^2$

M1: Use of constant acceleration equation with

$s = 0.196$, $u = 0$ and $a = 2.45$ to find t .

A1: Correct equation.

M1 A1

$$t = \sqrt{\frac{2 \times 0.196}{2.45}} = 0.4 \text{ seconds}$$

A1: Correct t

A1

3

(ii) $v^2 = 0^2 + 2 \times 2.45 \times 0.196$

$v = 0.98$

M1: Use of constant acceleration equation with

$s = 0.196$, $a = 2.45$, $u = 0$ and candidate's time

(as needed) to find v .

A1: Correct v .

M1A1

OR

$v = 0 + 2.45 \times 0.4 = 0.98 \text{ ms}^{-1}$

(M1A1)

OR

$0.96 = \frac{1}{2}(0 + v) \times 0.4$

(M1)

$v = 0.98 \text{ ms}^{-1}$

(A1)

2

[14]

Q3.

(a) $P \cos 80^\circ - Q \cos 80^\circ = 250a$

M1: Horizontal equation of motion in the form

$P \cos 80^\circ \pm Q \cos 80^\circ = 250a$ or

$P \sin 80^\circ \pm Q \sin 80^\circ = 250a$

A1: Correct horizontal equation.

M1A1

$P \sin 80^\circ + Q \sin 80^\circ = 250g$

B1: Correct vertical equation.

Note: the above marks could be awarded for a correct vector equation.

B1

$$P - Q = \frac{250a}{\cos 80^\circ}$$

$$P + Q = \frac{250g}{\sin 80^\circ} \quad \text{AG}$$

$$2P = \frac{250a}{\cos 80^\circ} + \frac{250g}{\sin 80^\circ}$$

dM1: Solving for P with an attempt to eliminate Q.

dM1

$$P = 125 \left(\frac{a}{\cos 80^\circ} + \frac{g}{\sin 80^\circ} \right)$$

A1: Correct result from correct working.

Must see an expression for 2P or 2P sin 80° cos 80°

A1

5

(b) $P \cos 80^\circ = 250a$

M1: Using Q = 0 into correct original equation(s) or resolving without Q.

M1

$P \sin 80^\circ = 250g$

$$\frac{1}{\tan 80^\circ} = \frac{a}{g}$$

dM1: Eliminating P

dM1

$$a = \frac{g}{\tan 80^\circ} = 1.73$$

A1: Correct a.

Note: use of $P = \pm Q$ scores M0dM0A0

Note: use of $P = 0$ can lead to ± 1.73 but scores M0dM0A0 unless fully justified by a symmetry argument.

A1

3

[8]

Q4.

(a) $a = \frac{dv}{dt}$

$a = -8e^{-2t} \mathbf{i} + (6 - 6t)\mathbf{j}$

M1: Differentiating with either of the two components correct. Do not need to see i or j .

A1: Correct i component.

A1: Correct j component.

M1

A1

A1

3

(b) (i) Using $F = ma$
 $F = 5 \times \{-8e^{-2t} \mathbf{i} + (6 - 6t)\mathbf{j}\}$

$= -40e^{-2t} \mathbf{i} + (30 - 30t)\mathbf{j}$

M1: Multiplying their acceleration by 5, even if not a vector.

A1: Correct expression.

M1

A1

2

(ii) Magnitude of F is

$\{(-40)^2 + (30)^2\}^{\frac{1}{2}}$

M1: Finding magnitude from two non-zero terms. Must add terms and square root. Condone $\{(40)^2 + (30)^2\}^{\frac{1}{2}}$

M1

$= 50$

A1: Correct answer only.

In this part, condone lack of negative signs in expression for force in (b) (i).

A1

2

(c) When F acts due west, j component is zero

$30 - 30t = 0$

M1: Putting j component equal to zero.

M1

$$t = 1$$

A1: Correct time.

A1

2

(d) $\mathbf{r} = -2e^{-2t} \mathbf{i} + (3t^2 - t^3) \mathbf{j} + \mathbf{c}$

M1: Integration with either of the two components correct.
Do not need to see $\mathbf{i}$ or $\mathbf{j}$.

A1: Correct $\mathbf{i}$ component.

A1: Correct $\mathbf{j}$ component.

Condone lack of $+\mathbf{c}$

M1

A1

A1

When $t = 0$, $\mathbf{r} = 6\mathbf{i} + 5\mathbf{j} \therefore \mathbf{c} = 8\mathbf{i} + 5\mathbf{j}$

dM1: Finding $\mathbf{c}$ using $6\mathbf{i} + 5\mathbf{j}$ and $e^0 = 1$.

dM1

$$\therefore \mathbf{r} = (8 - 2e^{-2t}) \mathbf{i} + (5 + \frac{1}{4}t^4 - t^3) \mathbf{j}$$

A1: Correct position vector.

A1

5

[14]

Q5.

(a) Using $F = ma$

$$-2mv^{\frac{5}{4}} = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -2v^{\frac{5}{4}}$$

AG

B1: Must see $-2mv^{\frac{5}{4}} = m \frac{dv}{dt}$ or $-2mv^{\frac{5}{4}} = ma$ and correct final answer.

B1

1

(b) $\int \frac{dv}{v^{\frac{5}{4}}} = -2 \int dt$

M1: Two integrals with one in the form $\int f(v)dv$ where $f(v) = v^{\pm \frac{5}{4}}$ or $v^{\pm \frac{4}{5}}$. The other integral must not contain v terms.

M1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t + c$$

A1: Correct expression.

Condone lack of + c for this A1, but no subsequent marks if no c.

A1

When $t = 0$, $v = 16 \Rightarrow c = -2$

dM1: Using $t = 0$ and $v = 16$ to find c.

A1: Obtaining $c = -2$.

dM1
A1

$$-\frac{4}{v^{\frac{1}{4}}} = -2t - 2$$

$$v^{\frac{1}{4}} = \frac{2}{1+t}$$

$$v = \left(\frac{2}{1+t}\right)^4$$

AG

$$v^{\frac{1}{4}} = \frac{2}{1+t}$$

A1: Correct final answer. Must see

or $v^{-\frac{1}{4}} = \frac{1+t}{2}$ or $\frac{1}{v^{\frac{1}{4}}} = \frac{1+t}{2}$
Or

if they obtain $v = \left(\frac{2}{1+t}\right)^4$

$v = 16, t = 0 \Rightarrow 16^{\frac{1}{4}} = \frac{2}{c}$, condone $c = 1$ (no other root considered)

A1

5

[6]

Q6.

(a) $\mathbf{v} = 4.5\mathbf{i} + 2\mathbf{j}$

B1: Correct velocity

B1

1

(b) $\mathbf{v} = 4.5\mathbf{j}$

B1: Correct velocity

B1
1

(c) Due North

B1: Correct direction

B1
1

(d) $4.5\mathbf{j} = 4.5\mathbf{i} + 2\mathbf{j} + 5\mathbf{a}$

M1: Use of a constant acceleration equation to find $\mathbf{a}$

A1: Correct equation

M1A1

$$\mathbf{a} = \frac{-4.5\mathbf{i} + 2.5\mathbf{j}}{5} = -0.9\mathbf{i} + 0.5\mathbf{j}$$

A1: Correct acceleration

A1
3

(e) $\mathbf{F} = 250(-0.9\mathbf{i} + 0.5\mathbf{j}) = -225\mathbf{i} + 125\mathbf{j}$

M1: Using Newton's second law.

M1

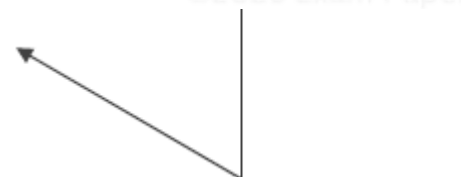
$$F = \sqrt{225^2 + 125^2} = 257 \text{ N}$$

M1: Finding magnitude.

A1: Correct magnitude of the force

M1A1
3

(f)



B1: Diagram with arrow in the correct quadrant.

B1

$$\tan^{-1}\left(\frac{125}{225}\right) = 29.1^\circ$$

M1: Use of trig to find an angle.

M1

A1: Correct angle from working.

A1

OR

$$\tan^{-1}\left(\frac{0.5}{0.9}\right) = 29.1^\circ$$

$$180 - 29.1 = 151^\circ$$

A1: Finding required angle correctly.

A1

4

[13]

Q7.

(a) $v^2 = 0^2 + 2 \times 9.8 \times 15$

M1: Use of constant acceleration equation to find v with $u = 0$ and $a = \pm 9.8$.

M1

$$v^2 = 294$$

A1: Correct equation

A1

$$v = 17.1 \text{ m s}^{-1}$$

A1: Correct speed from correct working. Accept AWRT 17.1. Accept 17.15.

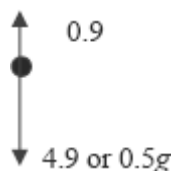
Accept $7\sqrt{6}$

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 17.2 m s^{-1} from $g = 9.81$.

A1

3

(b) (i)



B1: Correct diagram, with arrows and labels. Must see 0.9 and 4.9 or 0.5g (or 4.905 if working with $g = 9.81$).

B1

1

(ii) $4.9 - 0.9 = 0.5a$

M1: Uses $0.5a$.

B1: Explicit statement of " $4.9 - 0.9$ " or " $mg - 0.9$ " or " $0.5g - 0.9$ ".

M1B1

$(a =) \frac{4}{0.5} = 8 \text{ ms}^{-2}$ **AG**

A1: Correct acceleration from correct working. Can be awarded without the B1 mark.

Must see $\frac{4.9(\text{or } 0.5g) - 0.9}{0.5}$ or $\frac{4}{0.5}$ or $4 = 0.5a$

A1

Note: If $g = 9.81$ is used candidates will get 8.01 m s^{-2} . Deduct 1 mark if 8.01 is seen.

Examples:

$4.9 = 0.5a + 0.9$ M1 B0 A0

$a = 8$

$4 = 0.5a$ M1 B0 A1

$a = 8$

If candidates only write

$a = \frac{4}{0.5} = 8$ award M0 B0 A0.

3

(iii) $v^2 = 0^2 + 2 \times 8 \times 15$

M1: Use of constant acceleration equation to find v with $u = 0$ and $a = \pm 8$.

M1

$v = 15.5 \text{ m s}^{-1}$

A1: Correct speed from correct working.

Accept AWRT 15.5 or truncated to 15.4.

Accept $4\sqrt{15}$.

A1

2

- (iv) The air resistance force will not be constant, but changes **as the speed of the ball changes** (or **changes as the ball accelerates**).

B1: Correct explanation, key words in bold.

Do not award mark for statements that imply that the acceleration causes the air resistance to change.

B1

1

[10]

Q8.

(a) $(8\mathbf{i} + 12\mathbf{j}) + (4\mathbf{i} - 4\mathbf{j}) = 12\mathbf{i} + 8\mathbf{j}$

M1: Adding forces to find resultant.

A1: Correct resultant force.

M1A1

2

(b) $4\mathbf{a} = 12\mathbf{i} + 8\mathbf{j}$ or $(\mathbf{a} =) \frac{12\mathbf{i} + 8\mathbf{j}}{4}$

M1: Use of Newton's second law with 4a and their answer to part (a).

M1

$(\mathbf{a} =) 3\mathbf{i} + 2\mathbf{j}$ **AG**

A1: Correct acceleration from correct equation.

A1

2

(c) (i) $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + (3\mathbf{i} + 2\mathbf{j}) \times 20$
 $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + 60\mathbf{i} + 40\mathbf{j}$

B1: Seeing 60i + 40j or (3i + 2j) × 20

B1

M1: Use of constant acceleration equation with t = 20 and a = 3i ± 2j.

M1

$\mathbf{v} = (40\mathbf{i} + 32\mathbf{j}) - (60\mathbf{i} + 40\mathbf{j})$

AG

$= -20\mathbf{i} - 8\mathbf{j}$

*A1: Correct velocity from correct working, with one of the intermediate lines of working (or equivalent) shown.
 Note: Candidates may use u instead of v in their working.*

Example:

Starting with

$\mathbf{v} = 40\mathbf{i} + 32\mathbf{j} + (3\mathbf{i} + 2\mathbf{j}) \times 20$

Scores B1M1A0.

A1

3

Note on Verification Method:

$$\begin{aligned}
 \mathbf{v} &= (-20\mathbf{i} - 8\mathbf{j}) + (3\mathbf{i} + 2\mathbf{j}) \times 20 && \text{B1M1} \\
 &= (-20 + 60)\mathbf{i} + (-8 + 40)\mathbf{j} \\
 &= 40\mathbf{i} + 32\mathbf{j} && \text{A1}
 \end{aligned}$$

Similarly, verification to confirm acceleration from the two velocities is acceptable.

(ii) $(\mathbf{v} =) (-20\mathbf{i} - 8\mathbf{j}) + (3\mathbf{i} + 2\mathbf{j})$

B1: Correct velocity vector.

Note “v =” does not need to be seen.

B1

1

(iii) $(\mathbf{v} =) (3t - 20)\mathbf{i} + (2t - 8)\mathbf{j}$

M1: Velocity vector seen split into components. Condone omission of i and j

Note: This can be implied by later working, such as the second line of this solution.

M1

$$(3t - 20)^2 + (2t - 8)^2 = 8^2$$

dM1: Equation based on speed of 8.

dM1

A1: Correct unsimplified equation.

A1

$$13t^2 - 152t + 400 = 0$$

A1: Simplified quadratic equation

A1

$$t = \frac{152 \pm \sqrt{152^2 - 4 \times 13 \times 400}}{2 \times 13}$$

dM1: Solving quadratic equation, to obtain two solutions.

dM1

$$t = 4 \text{ or } t = 7.69$$

A1: Both correct solutions. Accept

$$\text{AWRT } 7.7 \text{ or } 7.6 \text{ or } \frac{100}{13}$$

Note: Using calculator to solve quadratic is acceptable.

A1

Q9.

(a)



B1: Force diagram with two arrows clearly in opposite directions. Must see 300 and one other label (a letter) or 550. Do not penalise if vertical forces included, even if wrong.

B1

1

(b) $T_1 - 300 = 500 \times 0.5$
 $(T_1 =) 300 + 250$

AG

$= 550 \text{ N}$

M1: Three term equation of motion.

M1

A1: Correct force from correct working.

Examples:

$T_1 = 300 + 250 = 550 \text{ N}$ scores M0A0

$T_1 - 300 = 250$ scores M1A1

$T_1 = 550 \text{ N}$

$T_1 = 300 + 500 \times 0.5 = 550 \text{ N}$ scores M1A1

Just $300 + 500 \times 0.5 = 550 \text{ N}$ scores M0A0

$700 + T_1 = 2500 \times 0.5$

$T_1 = 550$ scores M0A0

A1

2

(c) $T_2 - 550 - 300 = 500 \times 0.5$

M1: Four term equation of motion for Carriage 1 including 550 and 300 with mass 500

A1: Correct equation.

M1A1

$T_2 = 550 + 300 + 250 = 1100 \text{ N}$

A1: Correct force

A1

3

OR

$$T_2 - 600 = 1000 \times 0.5$$

M1: Three term equation of motion for Carriages 1 and 2 together including 300 twice or 600 with mass 1000.

(M1)

A1: Correct equation.

Accept $T_2 = 600 + 500$ or similar.

(A1)

$$T_2 = 600 + 500 = 1100 \text{ N}$$

A1: Correct tension

(A1)

3

(d) $P - 1100 - 400 = 2000 \times 0.5$

M1: Four term equation of motion for engine with mass 2000, a force of 400 and their answer to part (c).

M1

A1F: Correct equation.

A1F

$$P = 1100 + 400 + 1000 = 2500$$

A1F: Correct force.

A1F

3

OR

$$P - 1000 = 3000 \times 0.5$$

M1: Three term equation of motion for whole train with mass 3000 and 1000 (OE) force.

(M1)

A1F: Correct equation.

(A1F)

$$P = 1000 + 1500 = 2500$$

A1F: Correct force

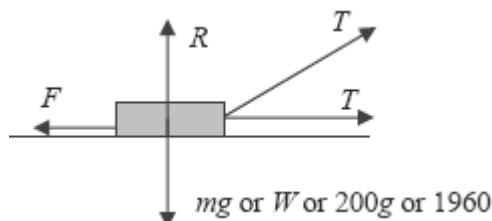
*Follow through from incorrect T_2 in part (c).
Don't penalise candidates who use a letter other than P .*

(A1F)

(3)

Q10.

(a)



B1: F , R and mg (or equivalent) with arrows and labels.

B1

*B1: Two **equal** tension forces with arrows and labels.*

B1

Ignore components if shown with a different notation, eg dotted lines.

2

(b) $R + T \sin 20^\circ = 1960$

OR

$R + T \sin 20^\circ = 200g$

M1: Resolving vertically with three terms. Must include $\sin 20^\circ$ or $\cos 20^\circ$ or $\sin 70^\circ$ or $\cos 70^\circ$ with T and $200g$ or 1960 .

A1: Correct equation.

M1A1

$(R =) 1960 - T \sin 20^\circ$

OR

$(R =) 200g - T \sin 20^\circ$

A1: Correct expression for the normal reaction.

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 1962 instead of 1960

A1

3

(c) $T \cos 20^\circ + T - F = 200 \times 0.3$

M1: Four term equation of motion. Must include $\sin 20^\circ$ or $\cos 20^\circ$ or $\sin 70^\circ$ or $\cos 70^\circ$ with T and a second T term with no trig.

A1: Correct equation

M1A1

$$T \cos 20^\circ + T - 0.4 (1960 - T \sin 20^\circ) = 200 \times 0.3$$

M1: Use of friction law with their expression for R, provided that R has two terms. Note that this mark does not depend on any previous marks.

Example:

If Candidate gives 1960 as answer to part (b), then: $F = 0.4 \times 1960 = 784$ scores M0 here

M1

$$T = \frac{60 + 784}{\cos 20^\circ + 1 + 0.4 \sin 20^\circ} = 406$$

dM1: Solving for T.

dM1

Note: This mark requires both of the previous M marks.

A1: Correct tension.

Accept AFWF 406 to 407.

Note: If $g = 9.81$ is used should get 407 instead of 406.

A1

5

[10]

Q11.

(a) Using $F = ma$,

$$-0.2mv^{\frac{1}{2}} = m \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

AG Must see equ'n containing m

B1

1

(b)
$$\int \frac{dv}{v^{\frac{1}{2}}} = - \int 0.2 dt$$

M1

$$2v^{\frac{1}{2}} = -0.2t + c$$

m1 for + c

A1m1

When $t = 0, v = 16 \therefore C = 8$

A1

$$2v^{\frac{1}{2}} = -0.2t + 8$$

$$v = (4 - 0.1t)^2$$

AG

A1

5

- (c) When $v = 1, 1 = (4 - 0.1t)^2$
 $4 - 0.1t = \pm 1$

M1

$t = 30$ or 50

$$\left[\text{if use } 2v^{\frac{1}{2}} = 8 - 0.2t \text{ no need to see } 50 \right]$$

A1

$t = 30$

$t \neq 50$ as ball stops when $t = 40$

A1

3

- (d) Integrating $v = (4 - 0.1t)^2$

$$v = 16 - 0.8t + 0.01t^2$$

$$x = 16t - 0.4t^2 + \frac{0.01}{3}t^3 + d$$

M1 for first 3 terms or $-\frac{10}{3}(4 - 0.1t)^3$

M1

When $t = 0, x = 0 \Rightarrow d = 0$

A1

$$x = 16t - 0.4t^2 + \frac{0.01}{3}t^3$$

When speed is $1 \text{ m s}^{-1}, t = 30$

$$x = 480 - 360 + 90$$

dep on M1 above

m1

$$= 210$$

[No 'd', 3 marks only]

A1

4

[13]

Q12.

(a) $12g - T = 12a$

M1: Three term equation of motion, with $12g$ (or 117.6), $12a$ (not $12ga$) and T .

A1: Correct equation

M1A1

$$T - 8g = 8a$$

M1: Three term equation of motion, with $8g$ (or 78.4), $8a$ (not $8ga$) and T .

A1: Correct equation

M1A1

$$4g = 20a$$

$$a \left(= \frac{4g}{20} \right) = 1.96 \text{ m s}^{-2} \quad \text{AG}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is -1.96 don't award final A1 mark.

Special Case:

Whole String Method $4g = 20a =$ and

$$a = \frac{4g}{20} = 1.96 \quad \text{OE M1A1A1}$$

5

(b) $T = 8g + 8 \times 1.96 = 94.1 \text{ N}$

M1: Use of three term equation of motion to find T , with $a = 1.96$.

A1: Correct tension.

Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v , with $a = 1.96$ and $u = 0$.

A1: Correct v .

Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v , with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

$v = 9.68 \text{ m s}^{-1}$

A1F: Correct v . FT initial velocity from (c)(i). For example 11.8 from 7.84.

A1F

3

Use of $g = 9.81$

(a) 1.962

M1A1M1A1A0

(b) 94.2

M1A1

(c) (ii) 9.69

M1A1A1

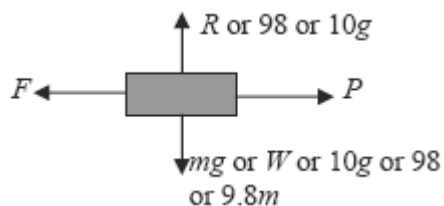
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[12]

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Q13.

(a)



B1: Correct force diagram with arrows and labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 10 kg for the weight.

Note: Do not accept μR or $0.5R$ for F .

B1

1

(b) (i) $(R = 10 \times 9.8 =) 98 \text{ N}$

*B1: Correct normal reaction. Accept 10g.
No need to see the letter R or working.*

B1

1

(ii) $(F \leq) 0.5 \times 98$

$(F \leq) 49$

*B1: Correct maximum value for friction.
Accept 5g.
No need to see the letter F or any
working. Ignore any inequalities.
For FT, must be 0.5 of candidate's answer to (b)(i).*

B1F

1

(iii) $(F =) 30 \text{ N}$

B1: Correct friction. Allow – 30.

B1

1

(c) $80 - 49 = 10a$

*M1: Three term equation motion,
containing 80, candidate's 49 and 10a
(not 10ga) in any combination.
A1F: Correct equation including signs.*

M1A1F

$a = 3.1 \text{ m s}^{-2}$

*A1F: Correct acceleration.
FT candidate's answer to (b)(ii).*

A1F

3

Allow use of $g = 9.81$

<i>(b) (i) 98.1</i>	<i>B1</i>
<i>(b) (ii) 49.05 or 49.1 or 49</i>	<i>B1</i>
<i>(c) 3.095 or 3.09 or 3.1</i>	<i>M1A1A1</i>

[7]

Q14.

(a) $12g - T = 12a$

*M1: Three term equation of motion, with
12g (or 117.6), 12a (not 12ga) and T.
A1: Correct equation*

M1A1

$$T - 8g = 8a$$

M1: Three term equation of motion, with $8g$ (or 78.4), $8a$ (not $8ga$) and T .

A1: Correct equation

M1A1

$$4g = 20a$$

$$a \left(= \frac{4g}{20} \right) = 1.96 \text{ m s}^{-2} \quad \text{AG}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is -1.96 don't award final A1 mark.

Special Case:

Whole String Method $4g = 20a$ and

$$a = \frac{4g}{20} = 1.96 \quad \text{OE M1A1A1}$$

5

(b) $T = 8g + 8 \times 1.96 = 94.1 \text{ N}$

M1: Use of three term equation of motion to find T , with $a = 1.96$.

A1: Correct tension.

Accept 94.08.

M1A1

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2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v , with $a = 1.96$ and $u = 0$.

A1: Correct v .

Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v , with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity

from (c)(i).

A1F

$$v = 9.68 \text{ m s}^{-1}$$

A1F: Correct v . FT initial velocity from (c)(i).
For example 11.8 from 7.84.

A1F

3

(iii) $4 = \frac{1}{2}(-3.92 + 9.68)t$

M1: Use of $s = \frac{1}{2}(u + v)t$

M1

A1: Correct values.

A1

A1: Correct signs.

A1

$$t = 1.39$$

dM1: Solving for t .

dM1

A1: Correct t .

A1

OR

$$-4 = 3.92t - 4.9t^2$$

$$4.9t^2 - 3.92t - 4 = 0$$

$$t = \frac{3.92 \pm \sqrt{3.92^2 - 4 \times 4.9 \times (-4)}}{2 \times 4.9}$$

M1: Forming a quadratic with candidates
 u from (c)(i) or v from (c)(ii) with 4.9 or 9.8.

(M1)

A1: Correct terms in quadratic.

(A1)

A1: Correct signs in quadratic.

(A1)

$$t = 1.39 \text{ or } t = -0.588$$

dM1: Solving quadratic (do not penalise for negative discriminant).

(dM1)

$$t = 1.39$$

A1: Correct root seen (other root does not need to be seen).

(A1)

OR

$$t_{\text{up}} + t_{\text{down}} = 0.4 + 0.4 + 0.588$$

M1: Finding total time from two or three times.

(M1)

A1: 0.4 or 0.8 seen.

(A1)

dM1: Finding second or third time for downward motion.

(dM1)

A1: Obtaining 0.588 or 0.988.

(A1)

$$= 1.39 \text{ (to 3SF)}$$

A1: 1.39. Accept 1.38.

(A1)

OR

$$9.68 = -3.92 + 9.8t$$

M1: Use of $v = u + at$

(M1)

A1: Correct values.

(A1)

A1: Correct signs.

(A1)

$$t = \frac{13.6}{9.8} = 1.39$$

dM1: Solving for t

(dM1)

A1: Correct t

(A1)

5

Use of $g = 9.81$

- | | |
|----------------|-------------|
| (a) 1.962 | M1A1M1A1A0 |
| (b) 94.2 | M1A1 |
| (c) (ii) 9.69 | M1A1A1 |
| (c) (iii) 1.39 | M1A1A1dM1A1 |

[17]

Q15.

(a) $10\mathbf{a} = 9\mathbf{i} + 12\mathbf{j}$

M1: Application of Newton's second Law with $m = 10$ in vector form.

M1

$\mathbf{a} = (0.9\mathbf{i} + 1.2\mathbf{j}) \text{ m s}^{-2}$

*A1: Correct acceleration.
If acceleration incorrect follow their value through for the rest of this question.*

A1

2

(b) (i) $\mathbf{r}(5) = (2.2\mathbf{i} + 1\mathbf{j}) \times 5 + \frac{1}{2} (0.9\mathbf{i} + 1.2\mathbf{j}) \times 5^2$

M1: Use of constant acceleration to find position vector at $t = 5$, with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$.

M1

$= 22.25\mathbf{i} + 20\mathbf{j}$

A1F: Correct position vector, for candidate's acceleration which must be a vector. Allow $22.3\mathbf{i} + 20\mathbf{j}$.

A1F

$d = \sqrt{22.25^2 + 20^2} = 29.9 \text{ metres}$

dM1: Calculation of distance from position vector. Must see + sign.

dM1

*A1F: Correct distance, for their acceleration.
Accept 30 from $22.3\mathbf{i} + 20\mathbf{j}$.*

A1F

4

(ii) $\mathbf{v} = (2.2\mathbf{i} + 1\mathbf{j}) + (0.9\mathbf{i} + 1.2\mathbf{j})t$

M1: Use of constant acceleration equation to find an expression for $\mathbf{v}$, with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$

M1

A1F: Correct $\mathbf{v}$ for their acceleration.

A1F

2

(iii) $\mathbf{v} = (2.2 + 0.9t)\mathbf{i} + (1 + 1.2t)\mathbf{j}$
 $2.2 + 0.9t = 1 + 1.2t$
 $1.2 = 0.3t$
 $t = 4$

M1: Equation involving both $\mathbf{i}$ and $\mathbf{j}$ components of their velocity. Could have incorrect signs, for example $2.2 + 0.9t = -(1 + 1.2t)$.

M1

A1F: Correct equation.

A1F

A1F: Correct time, for their acceleration.

A1F

3

[11]

Q16.

(a) Using $\mathbf{F} = m\mathbf{a}$,

$$400 \cos \frac{\pi}{2} t \mathbf{i} + 600t^2 \mathbf{j} = 200 \mathbf{a}$$

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$$\mathbf{a} = 2 \cos \frac{\pi}{2} t \mathbf{i} + 3t^2 \mathbf{j}$$

A1

2

(b) $\mathbf{v} = \int \mathbf{a} \, dt$

M1 for either $\int \mathbf{a} \, dt$ or 1 of 2 terms correct

M1

$$= \frac{4}{\pi} \sin \frac{\pi}{2} t \mathbf{i} + t^3 \mathbf{j} + \mathbf{c}$$

$m1$ for + c

A1m1

When $t = 4$, $\mathbf{r} = -3\mathbf{i} + 56\mathbf{j}$,
 $64\mathbf{j} + \mathbf{c} = -3\mathbf{i} + 56\mathbf{j}$

m1

$$\therefore \mathbf{c} = -3\mathbf{i} - 8\mathbf{j}$$

$$\therefore \mathbf{v} = \left(\frac{4}{\pi} \sin \frac{\pi}{2} t - 3\right)\mathbf{i} + (t^3 - 8)\mathbf{j}$$

$\frac{2}{\pi}$
 Do not accept $\frac{2}{\pi}$ Accept 1.27 for $\frac{4}{\pi}$

A1

5

- (c) When particle is moving due west, northerly component is zero

M1

$$\therefore t^3 - 8 = 0$$

A1ft

$$t = 2$$

A1

3

- (d) When $t = 2$, $\mathbf{v} = -3\mathbf{i} + 0\mathbf{j}$

B1ft

Speed of particle is 3 m s^{-1}

$B1$ for change -3 to $+3$

B1

2

[12]

Q17.

- (a) $11g - T = 11a$

M1: Equation of motion for A, containing T, 11g or 107.8 and 11a.

M1

A1: Correct equation

A1

$$T - 9g = 9a$$

M1: Equation of motion for B containing T, 9g or 88.2 and 9a.

M1

A1: Correct equation

A1

$$2g = 20a$$

AG

$$a = 0.98 \text{ m s}^{-2}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and $a = 2g / 20 = 0.98$ OE M1A1A1

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(b) (i) $v = 0 + 0.98 \times 0.5 = 0.49 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225 \text{ m}$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$$0.49^2 = 0^2 + 2 \times 0.98s$$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and their v .

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$$d = 2 \times 0.1225$$

M1: Doubling distance or use of $d/2$ in their original equation.

M1

$$= 0.245 \text{ m}$$

*A1: Correct final distance. Allow 0.244 or 0.246.
(Use of $0.5 \times 0.49 = 0.245$ scores zero unless justified)*

A1

4

[11]

Q18.

(a) $4a = 4g \sin 40^\circ$

M1: Resolving and application of Newton's second law. Allow $\cos 40^\circ$

M1

A1: Correct expression.

A1

$$a = g \sin 40^\circ = 6.30 \text{ m s}^{-2} \quad \mathbf{AG}$$

*A1: Correct result from correct working.
Must see 6.30 not 6.3.*

Just seeing $g \sin 40^\circ = 6.30 \text{ m s}^{-2}$ scores full marks.

Use of $g = 9.81$ gives 6.31, M1A1A0, but don't penalise again on the same script.

A1

3

(b) $0.9 = 0 + \frac{1}{2} \times a \times 0.6^2$

M1: Use of a constant acceleration equation to find a , with $s = 0.9$, $u = 0$ and $t = 0.6$.

M1

A1: Correct equation

A1

$$a = \frac{0.9 \times 2}{0.6^2} = 5 \text{ m s}^{-2}$$

A1: Correct acceleration

A1

ALT Method

$$0.9 = \frac{1}{2} (0 + v) \times 0.6$$

$$v = 3$$

No marks at this stage.

$$3 = 0 + 0.6a$$

M1: Constant acceleration equation with $u = 0$ and $t = 0.6$.

(M1A1)

A1: Correct equation

$$a = 5 \text{ m s}^{-2}$$

A1: Correct acceleration.

(A1)

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3

- (c) The acceleration is reduced because of air resistance or the fact that there is friction.

B1: Must mention air resistance/resistive forces or friction. Do not allow air friction.

B1

1

[7]

Q19.

- (a) Peg is smooth

B1: Correct assumption

B1

1

(b) String is light

B1: First correct assumption

B1

String is inextensible or inelastic

B1: Second correct assumption

B1

Tension is the same throughout the string

Note: Ignore any additional assumptions.

2

(c) $11g - T = 11a$

M1: Equation of motion for A, containing, 11g or 107.8 and 11a.

M1

A1: Correct equation

A1

$$T - 9g = 9a$$

M1: Equation of motion for B containing, 9g or 88.2 and 9a.

M1

A1: Correct equation

A1

$$2g = 20a$$

AG

$$a = 0.98 \text{ m s}^{-2}$$

A1: Correct acceleration from correct orking.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is – 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and

$$a = 2g/20 = 0.98 \text{ OE M1A1A1}$$

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(d) (i) $v = 0 + 0.98 \times 0.5 = 0.49 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225 \text{ m}$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$$0.49^2 = 0^2 + 2 \times 0.98s$$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and their v

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$$d = 2 \times 0.1225$$

M1: Doubling distance or use of $d/2$ in their original equation.

M1

$$= 0.245 \text{ m}$$

A1: Correct final distance. Allow 0.244 or 0.246. (Use of $0.5 \times 0.49 = 0.245$ scores zero unless justified)

A1

4

If candidates calculate the distance first award marks as above (see (d) (i)) or:

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $s = 0.1225$.

A1: Correct v

Note: If parts (i) and (ii) are not separated or clearly labelled still award marks for both parts if justified.

[14]

Q20.

(a) Resultant = $(6\mathbf{i} - 3\mathbf{j}) + (3\mathbf{i} + 15\mathbf{j})$
 $= 9\mathbf{i} + 12\mathbf{j}$

M1: Summing the two vectors

M1

A1: Correct resultant

A1

2

(b) Magnitude = $\sqrt{9^2 + 12^2}$
 $= 15 \text{ N}$

M1: Finding magnitude with an addition sign.

M1

A1F: Correct magnitude based on their answer to part (a).

A1F

2

(c) $1.5m = 9$ $2m = 12$
 or
 $m = 6 \text{ kg}$ $m = 6 \text{ kg}$

M1: Applying Newton's second law to one or both of the components.

M1

A1F: Correct mass, follow through their answer to part (a). Do not award this mark if vector division with 2 components has been used, eg

$\frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6$ or $6\mathbf{i} + 6\mathbf{j}$ etc without a correct previous statement gives M0A0

A1F

2

(d) (i) $\mathbf{r} = \frac{1}{2} (1.5\mathbf{i} + 2\mathbf{j})t^2$

M1: Using a constant acceleration equation to find the position vector with $u = 0i + 0j$

M1

A1: Correct position vector.

A1

2

(ii)
$$\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j}) \times 2^2 = 3\mathbf{i} + 4\mathbf{j}$$

$$d = \sqrt{(3)^2 + (4)^2}$$

M1: Finding the position vector when $t = 2$

($\mathbf{r} = (1.5\mathbf{i} + 2\mathbf{j}) \times 2 = 3\mathbf{i} + 4\mathbf{j}$ scores M0 unless it is clear how the 2 was obtained, possibly by a correct formula in (d) (i))

M1

$$= \sqrt{25} = 5$$

A1: Correct distance

A1

2

[10]

Q21.

(a)
$$\mathbf{v} = \frac{d\mathbf{r}}{dt}$$

M1

$$\mathbf{v} = (e^{\frac{1}{2}t} - 8)\mathbf{i} + (2t - 6)\mathbf{j}$$

i terms

A1

j terms

A1

3

(b) (i) When $t = 3$, $\mathbf{v} = -3.52\mathbf{i}$

Accept $(e^{\frac{3}{2}} - 8)\mathbf{i}$

B1

Speed is 3.52 m s^{-1}

3.5 does not give 2nd B mark

B1

2

(ii) West

B1

1

(c) $\mathbf{a} = \frac{1}{2}e^{\frac{1}{2}t}\mathbf{i} + 2\mathbf{j}$

M1A1

When $t = 3$, $\mathbf{a} = \frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}$ or $2.24\mathbf{i} + 2\mathbf{j}$

A1

3

(d) Using $\mathbf{F} = m\mathbf{a}$:

Accept $\mathbf{F} = 7\mathbf{a}$

M1

$$\mathbf{F} = 7 \left(\frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j} \right)$$

$\therefore$ Magnitude of force is $7 \left(\left(\frac{1}{2}e^{\frac{3}{2}} \right)^2 + 2^2 \right)^{\frac{1}{2}}$

M1

$\mathbf{F} = 21.025$

$\mathbf{F} = 21.0$

Accept 21

A1

3

[12]

Q22.

(a) Using $F = ma$:

$$-0.08v^2 = 0.05a$$

B1

$$\therefore \frac{dv}{dt} = -1.6v^2$$

AG; condone sign error in first B1

B1

2

(b) $\int \frac{dv}{v^2} = -1.6 \int dt$

M1

$$-\frac{1}{v} = -1.6t(+c)$$

Condone $-\frac{1}{v} = -1.6t + c \Rightarrow \frac{1}{v} = 1.6t + c$

A1

When $t = 0, v = 3 \Rightarrow c = -\frac{1}{3}$

M1

$$\frac{1}{v} = \frac{1}{3} + 1.6t \quad *$$

A1

$$\frac{1}{v} = \frac{1}{3} + \frac{8}{5}t$$

$$\frac{1}{v} = \frac{5 + 24t}{15}$$

$$v = \frac{15}{5 + 24t}$$

AG; all working lines correct from *

A1

5

[7]

Q23.

- (a) Resultant Force = 300×2.2
 = 660 N AG

B1: Correct value from correct multiplication.

B1
1

(b) $P - 400 = 660$
 $P = 1060$

M1: Three term equation of motion

M1

A1: Correct value for P

A1
2

(c) $23 = 12 + 2.2t$

M1: Use of a constant acceleration equation to find t.

M1

A1: Correct equation

A1

$$t = \frac{23 - 12}{2.2} = 5 \text{ s}$$

A1: Correct time

A1
3

[6]

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Q24.

(a) $13^2 = 0^2 + 2 \times 1.3s$

M1: Use of a constant acceleration equation to find distance.

M1

A1: Correct equation

A1

$$s = \frac{13^2}{2.6} = 65 \text{ m}$$

A1: Correct distance

A1
3

- (b) (i) $3900 - 800 - P = 2000 \times 1.3$
M1: Four term equation of motion for car and trailer.

M1

A1: Correct equation

A1

$$P = 3900 - 800 - 2600 = 500 \text{ N}$$

A1: Correct value for P

A1

3

- (ii) $T - 500 = 600 \times 1.3$
M1: Three term equation of motion for trailer.

M1

A1: Correct equation

A1F

$$T = 500 + 780 = 1280 \text{ N}$$

A1: Correct tension

A1F

3

[9]

Q25.

- (a) Resultant Force = $3000 - 600 = 2400 \text{ N}$
M1: Difference between the two forces.

M1

*A1: Correct magnitude of resultant force.
 Must be a positive answer.
 (600 - 3000 = -2400 scores M1A0)*

A1

2

- (b) $2400 = 1200a$
M1: Use of Newton's second Law to find acceleration.

M1

$$a = \frac{2400}{1200} = 2 \text{ m s}^{-2}$$

A1: Correct acceleration

$$(a = \frac{-2400}{1200} = -2 \text{ m s}^{-2} \text{ scores M1 A0})$$

A1

2

[4]

Q26.

(a) $R = 14 \times 9.8 = (137.2)$

M1: Finding the normal reaction. Accept 14g.

M1

$$F = 0.25 \times 137.2 \text{ OR } F = 0.25 \times 14 \times 9.8$$

M1: Use of $F = \mu R$

M1

$$F = 34.3 \text{ N}$$

A1: Correct friction

Use of $g = 9.81$ gives

$R = 137.3$ and $F = 34.3$ so in this case do not penalise use of $g = 9.81$.

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A1

3

(b) $6g - T = 6a$ ©2025 Exam Papers Practice. All rights reserved.

M1: Equation of motion for the particle, containing T , $6g$ or 58.8 and $6a$.

A1: Correct equation with correct signs.

M1 A1

$$T - 34.3 = 14a$$

M1: Equation of motion for the block, containing T , 34.3 or their F and $14a$.

A1: Correct equation with correct signs.

M1 A1

$$6g - 34.3 = 20a$$

$$a = \frac{6g - 34.3}{20} = 1.225 \text{ m s}^{-2}$$

AG

*A1: Correct acceleration from correct working.
If – 1.225 is obtained from consistent working award 4 marks and if changed to +1.225 with an explanation, award full marks.*

Special Case:

Whole string method

$$6g - 34.3 = 20a$$

$$a = 1.225 \text{ OE}$$

award M1A1A1

Use of $g = 9.81$ gives

$$a = 1.228 \text{ penalise use of}$$

$g = 9.81$ by deducting 1 mark, but don't penalise again on the same script.

(c) $T - 34.3 = 14 \times 1.225$

M1: Use of either of candidates equations of motion to find tension, with $a = \pm 1.225$ and their F (Method 1).

M1

$$T = 17.15 + 34.3 = 51.5 \text{ N}$$

A1: Correct tension

Accept 51.45 or 51.4. Don't penalise use of $g = 9.81$ if already done in part (b).

A1

OR

$$6g - T = 6 \times 1.225$$

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(M1)

$$T = 6 \times 9.8 - 6 \times 1.225 = 51.5$$

(A1)

2

(d) $v^2 = 0^2 + 2 \times 1.225 \times 0.8$

M1: Use of constant acceleration equation to find speed with $u = 0$.

A1: Correct equation

M1A1

$$v = \sqrt{1.96} = 1.4 \text{ m s}^{-1}$$

A1: Correct speed AWRT 1.4

A1

OR

$$0.8 = \frac{1}{2} \times 1.225t^2$$

In method 2, no marks awarded for just finding t .

$$t = (1.1428)$$

(M1)

$$v = 1.225 \times 1.1428$$

(A1)

$$= 1.40$$

(A1)

3

(e) $v^2 = 1.4^2 + 2 \times 9.8 \times 0.5$

M1: Use of constant acceleration equation to find speed with $u = 1.4$ or their answer to part (d), $a = \pm 9.8$ and $s = 0.5$.

A1F: Correct equation.

Follow through their answer to part (d).

M1A1F

$$v = 3.43 \text{ m s}^{-1}$$

A1F: Correct speed.

Don't penalise use of $g = 9.81$ if already done earlier in question.

In method 2, no marks awarded for just finding t .

A1F

OR

$$0.5 = 1.4t + 4.9t^2$$

$$t = 0.2071$$

$$v = 1.4 + 9.8 \times 0.2071$$

(M1)(A1F)

$$= 3.43 \text{ m s}^{-1}$$

(A1F)

3

Q27.

(a) (i) $20 \times 9.8 = R + 60 \sin 30^\circ$

M1: Equation or expression for normal reaction with mg or $20g$ or 196 and $60 \sin 30^\circ$ or $60 \cos 30^\circ$.

A1: Correct equation or expression with correct signs.

M1A1

$(R =) 20 \times 9.8 - 60 \sin 30^\circ = 166 \text{ N}$ AG

A1: Correct value from correct working. Must be positive.

Don't penalise use of $g = 9.81$ if already done earlier on script. Should still get 166, but from 166.2.

A1

3

(ii) $166\mu = 60 \cos 30^\circ$

M1: Use of $F = \mu R$, with $R = 166$ or 166.2 . Do not allow inequalities here.

M1: Resolving horizontally with $\cos 30^\circ$ or $\sin 30^\circ$ oe

A1: Correct equation

Examples:

$166\mu = 60$ M1M0A0

$166\mu = -60 \cos 30^\circ$ M1M1A0

M1M1A1

$$\mu = \frac{60 \cos 30^\circ}{166}$$

$= 0.313$

A1: Correct coefficient of friction.

A1

4

(b) $20 \times 0.8 = T \cos 30^\circ - 0.313(20 \times 9.8 - T \sin 30^\circ)$

B1: $20g - T \sin 30^\circ$ oe seen.

M1: Three term equation of motion, where normal reaction is dependent on T .

A1F: Correct equation

B1M1A1F

$$T = \frac{20 \times 0.8 + 0.313 \times 20 \times 9.8}{\cos 30^\circ + 0.313 \sin 30^\circ} = 75.6 \text{ N}$$

dM1: Solving for T including factorisation.

A1F: Correct tension.

AWRT 75.6

Follow through incorrect values of μ from part (a).

Don't penalise use of $g = 9.81$ if already done earlier on script. Should get 75.7.

Allow 75.8 if intermediate values rounded.

5

[12]

Q28.

(a) $\mathbf{a} = \frac{d\mathbf{v}}{dt} = (3t^2 - 15)\mathbf{i} + (6 - 2t)\mathbf{j}$

A1 (i terms) A1 (j terms)

M1A1A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$:
Force = $4 \times \{(3t^2 - 15)\mathbf{i} + (6 - 2t)\mathbf{j}\}$

M1

$$= (12t^2 - 60)\mathbf{i} + (24 - 8t)\mathbf{j}$$

AG

A1

2

(ii) When $t = 2$, force = $-12\mathbf{i} + 8\mathbf{j}$

M1A1

$$\text{Magnitude of force} = \sqrt{12^2 + 8^2} \text{ N}$$

M1

$$= 14.4 \text{ (N)}$$

A1

4

[9]

Q29.

(a) Using $F = ma$:

$$-\lambda m v^{\frac{3}{2}} = m \frac{dv}{dt}$$

M1

$$\therefore \frac{dv}{dt} = -\lambda v^{\frac{3}{2}}$$

AG

A1

2

(b) $\int \frac{dv}{v^{\frac{3}{2}}} = -\lambda \int dt$

M1

$$-\frac{2}{v^{\frac{1}{2}}} = -\lambda t + c$$

Condone no '+c'

A1

When $t = 0, v = 9 \Rightarrow c = -\frac{2}{3}$

Dep. on correct integration

(accept sign or ' $\frac{1}{2}$ ' error)

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M1A1

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3}$$

A1

$$\frac{\sqrt{v}}{2} = \frac{1}{\lambda t + \frac{2}{3}}$$

$$v = \left(\frac{6}{2 + 3\lambda t} \right)^2$$

Needs correct algebra

m1

$$v = \frac{36}{(2 + 3\lambda t)^2}$$

AG

A1

7

(c) When $v = 4$,

$$\text{or } \frac{36}{(2 + 3\lambda t)^2} = 4 \quad \text{M1}$$

$$\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3} \Rightarrow 1 = \lambda t + \frac{2}{3}$$

$$(2 + 3\lambda t)^2 = 9 \quad \text{A1}$$

M1A1

$$t = \frac{1}{3\lambda}$$

$$t = \frac{1}{3\lambda} \quad \text{A1 needs statement why}$$

$$2 + 3\lambda t \neq -3$$

A1

3

[12]

Q30.

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$$8 = \frac{1}{2} a \times 5^2 \quad \text{©2025 Exam Papers Practice. All rights reserved.}$$

(a)

*Use of constant acceleration equation
with $u = 0$ to find a .*

M1

$$a = \frac{2 \times 8}{25} = 0.64 \text{ m s}^{-2}$$

AG

*Correct answer from correct working,
showing evidence of solving for a .
Allow verification / substitution.*

A1

2

(b) $T - 70 \times 9.8 = 70 \times 0.64$

Three term equation of motion for crate.

M1

Correct equation

A1

$$T = 730.8 = 731 \text{ N to 3 sf}$$

Correct tension

A1

3

(c) $v = \frac{8}{5} = 1.6 \text{ m s}^{-1}$

Correct average speed.

Accept $\frac{8}{5}$

Allow $\frac{3.2 + 0}{2} = 1.6 \text{ m s}^{-1}$

B1

1

[6]

Q31.

(a) (i) $0.2a = -0.2 \times 9.8 \sin 20^\circ$

AG

$$a = -9.8 \sin 20^\circ = -3.35 \text{ m s}^{-2}$$

Two term equation of motion with weight resolved

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M1

Correct equation

A1

Correct acceleration from correct working

A1

SC No negative sign but otherwise correct
award M1A1A0

Allow $a = -g \sin 20^\circ$

3

(ii) $0 = 4^2 + 2 \times (-3.35)s$

Use of constant acceleration equation
with $v = 0$ and $u = 4$

M1

Correct equation

A1

$$s = \frac{16}{6.7} = 2.39 \text{ m}$$

Correct distance

A1

3

- (iii) The puck slides back down the slope as the puck is at rest and the resultant force is now acting down the slope / no friction / smooth slope.

Slides back down

B1

Acceptable explanation

E1

2

(b) (i) $R = 0.2 \times 9.8 \cos 20^\circ$ **AG**

*Finding normal reaction by resolving.
Must see a trig term.*

M1

$$F = 0.5 \times 0.2 \times 9.8 \cos 20^\circ$$

Use of $F = \mu R$

M1

$$= 0.921 \text{ N}$$

Correct friction from correct working.

A1

3

(ii) $0.2a = -0.921 - 0.2 \times 9.8 \sin 20^\circ$

Three term equation of motion with the weight resolved

M1

Correct equation

A1

$$a = -7.96 \text{ m s}^{-2}$$

Correct acceleration (with or without the minus sign, applied to both A1 marks)

A1
3

- (iii) The puck stays at rest because the friction has a maximum of 0.921 and the component of the weight down the slope is less (0.670)

Stays at rest

B1

Acceptable explanation

dE1
2

[16]

Q32.

(a) $F = 0.4 \times 1000 \times 9.8$

Use of $F = \mu R$

M1

$= 3920$

AG

*Correct friction from correct working.
Allow $F = 0.4 \times 9800$
Allow verification*

A1
2

(b) $P - 3920 = 5000 \times 0.8$

Three term equation of motion including an explicit 0.8

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M1

Correct equation

A1

$P = 7920 \text{ N}$

AG

*Correct force from correct working.
Allow $P = 5000 \times 0.8 + 3920$*

A1
3

(c) $T - 3920 = 1000 \times 0.8$

Three term equation of motion

M1

Correct equation

A1

$$T = 4720 \text{ N}$$

Correct tension

A1

or

$$7920 - T = 4000 \times 0.8$$

$$T = 4720 \text{ N}$$

3

- (d) Friction is reduced because the normal reaction is reduced.

Friction reduced

B1

Acceptable explanation

E1

2

[10]

Q33.

(a) $\mathbf{v} = \frac{d\mathbf{r}}{dt}$

$$\mathbf{v} = (3t^2 - 6t)\mathbf{i} + (4 + 2t)\mathbf{j}$$

M1A1

2

(b) (i) $\mathbf{a} = (6t - 6)\mathbf{i} + 2\mathbf{j}$

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M1A1ft

Using $\mathbf{F} = m\mathbf{a}$:

$$\mathbf{F} = (18t - 18)\mathbf{i} + 6\mathbf{j}$$

A1ft

3

(ii) When $t = 3$, $\mathbf{F} = 36\mathbf{i} + 6\mathbf{j}$

Magnitude is $\sqrt{36^2 + 6^2}$

M1

$$= 36.5$$

Accept $6\sqrt{37}$; ft from (b)(i)

A1ft 2

- (c) When **F** acts due north:
Component of **F** in the **i** direction is 0

M1

$$18t - 18 = 0$$

$$t = 1$$

ft from (b)(i)

A1ft 2

[9]

Q34.

(a) $2 \times 9.8 - T = 2 \times 0.9$

Three term equation of motion for B

M1

Correct equation

A1

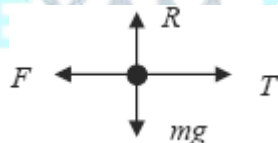
$$T = 19.6 - 1.8 = 17.8 \text{ N}$$

Correct tension

A1

3

- (b)



Correct force diagram with labels and arrows

B1

1

(c) $R = 3 \times 9.8 = 29.4 \text{ N}$

Calculation of normal reaction force.

B1

1

(d) $17.8 - F = 3 \times 0.9$

Three term equation of motion for A.

M1

$$F = 17.8 - 2.7 = 15.1 \text{ N}$$

Correct friction

A1

2

(e) $15.1 = 29.4\mu$

Use of $F = \mu R$

M1

$$\mu = \frac{15.1}{29.4} = 0.514$$

Correct value for μ

A1

2

[9]

Q35.

(a) $s = \frac{1}{2}(3 + 10) \times 3$

Finding distance by summing 3 areas or using formula for the area of a trapezium

M1

Correct equation/3 correct expressions for the areas

A1

$$= 19.5 \text{ m}$$

Correct total distance

A1

3

(b) $\alpha = \frac{3}{4} = 0.75 \text{ m s}^{-2}$

Correct acceleration as a decimal or as a fraction

B1

1

(c) $T - 400g = 400 \times 0.75$

Three term equation of motion containing
 T , $400g$ and 400×0.75 or equivalent

M1

Correct equation

A1F

$$T = 3920 + 300 = 4220 \text{ N}$$

Correct tension

A1F

Only ft from $\alpha = \frac{4}{3}$

(ft 4453 N or 4450 N from $\alpha = \frac{4}{3}$ scores M1A1A1)

3

[7]



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